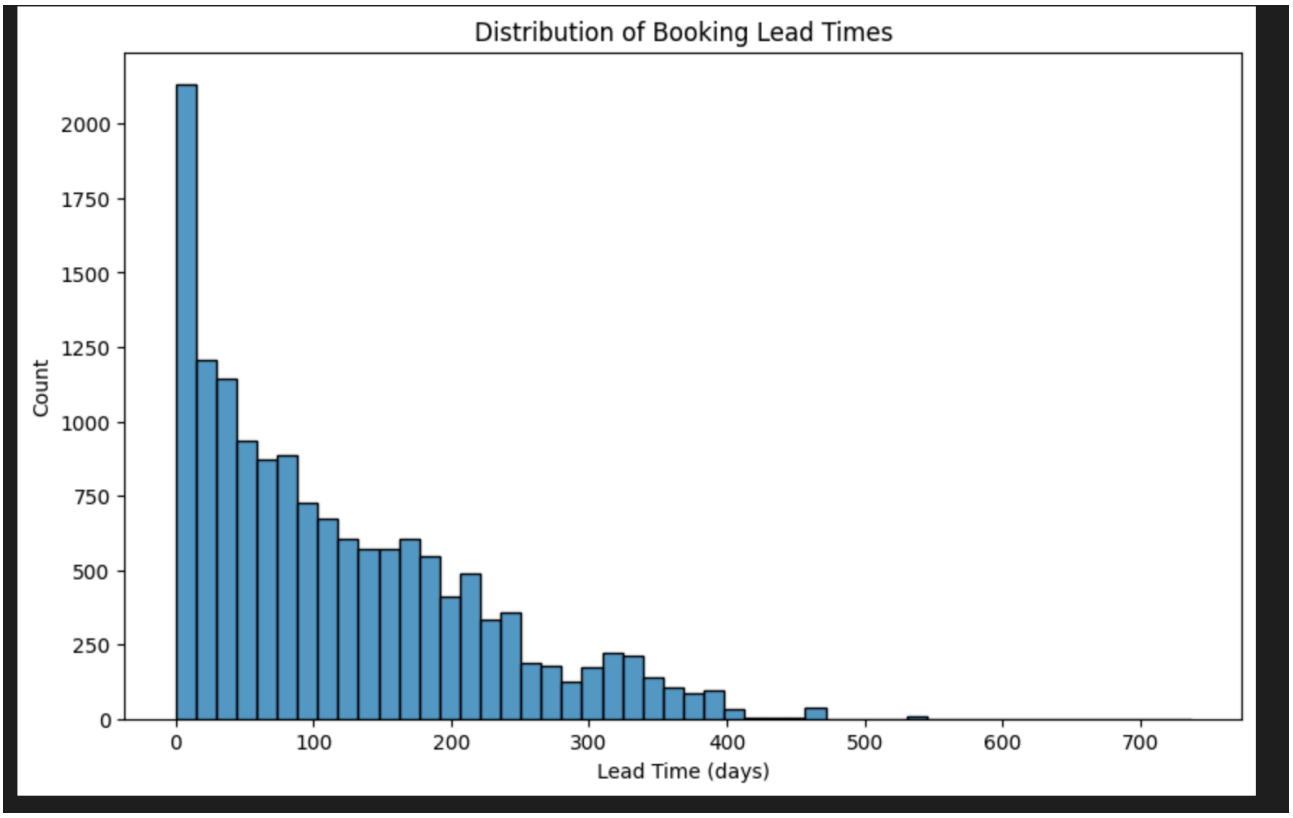
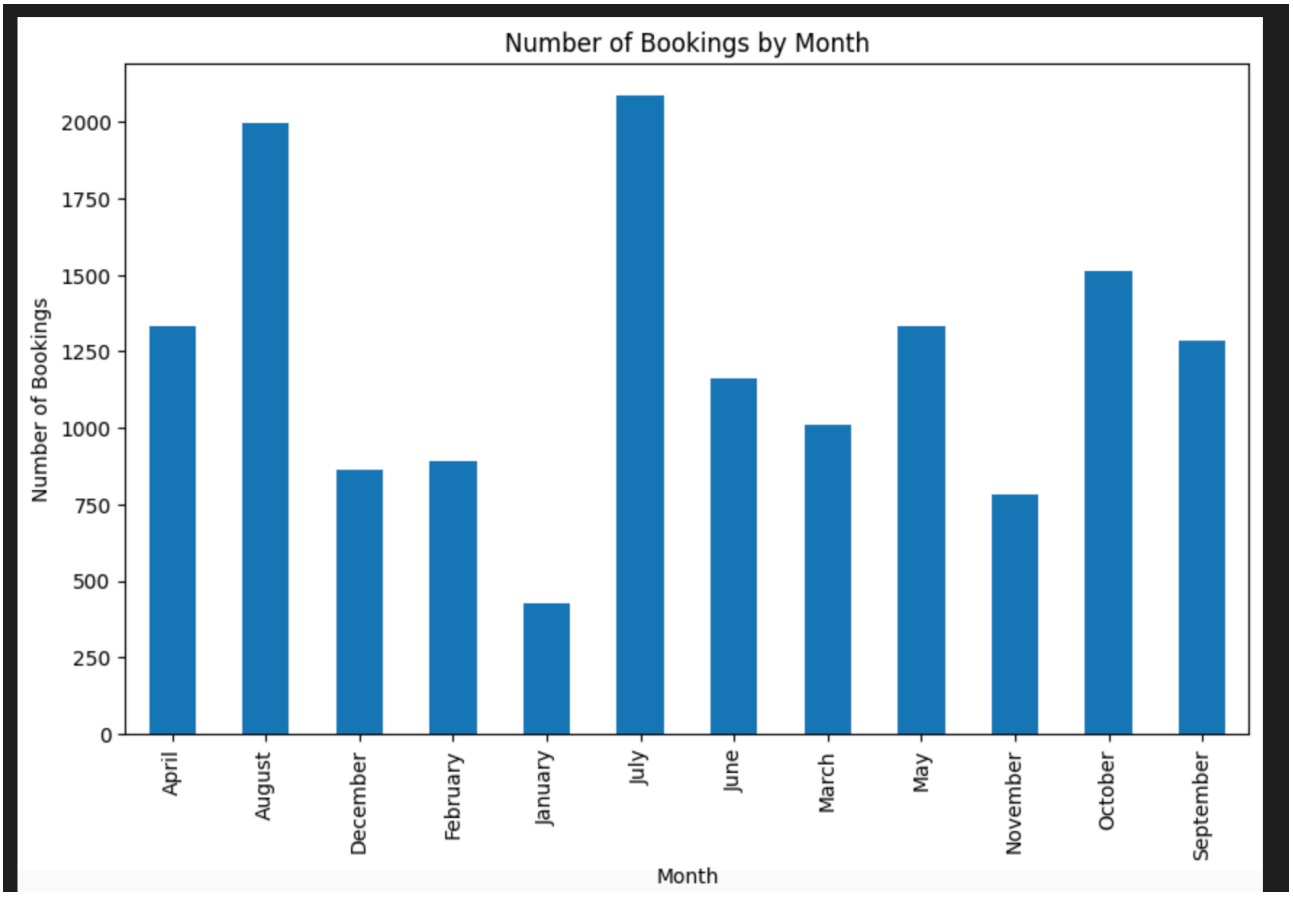
```python
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np

# Load the data (note that hotel_bookings.csv is present
in the same directory as the notebook)
df = pd.read_csv('hotel_bookings.csv')

# Display basic information about the dataset
print(df.info())
print(df.head())

# 1. Analyze booking timing
plt.figure(figsize=(10, 6))
sns.histplot(df['lead_time'], bins=50)
plt.title('Distribution of Booking Lead Times')
plt.xlabel('Lead Time (days)')
plt.ylabel('Count')
plt.show()

# Analyze bookings by month
monthly_bookings =
df['arrival_date_month'].value_counts().sort_index()
plt.figure(figsize=(10, 6))
monthly_bookings.plot(kind='bar')
plt.title('Number of Bookings by Month')
plt.xlabel('Month')
plt.ylabel('Number of Bookings')
plt.show()

# 2. Analyze length of stay and daily rates
df['total_nights'] = df['stays_in_weekend_nights'] +
df['stays_in_week_nights']
correlation = df[['total_nights', 'adr']].corr()
plt.figure(figsize=(10, 6))
sns.scatterplot(x='total_nights', y='adr', data=df)
plt.title('Length of Stay vs. Daily Rate')
plt.xlabel('Length of Stay (nights)')
plt.ylabel('Average Daily Rate')
plt.show()
```

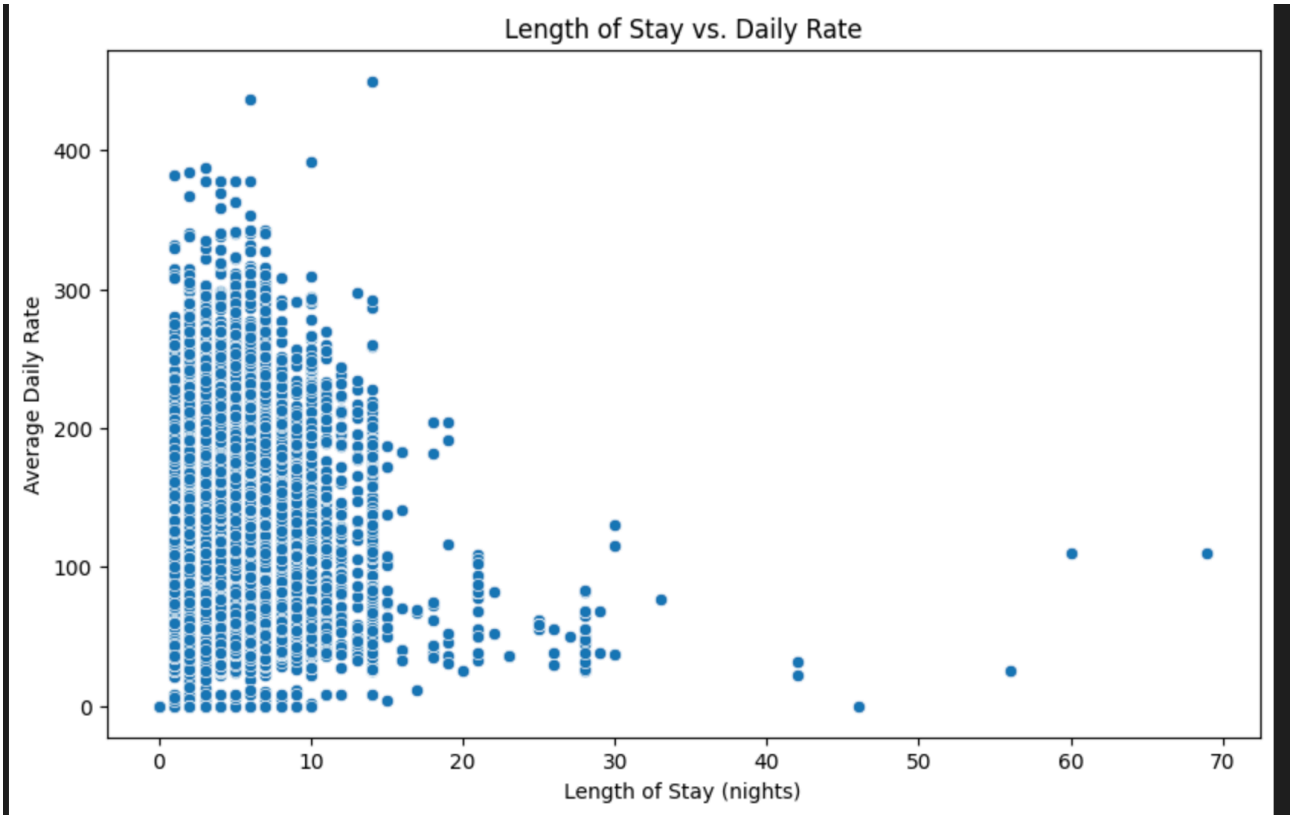
```
print(f"Correlation between length of stay and daily
rate: {correlation.iloc[0, 1]:.2f}")

# 3. Analyze demographic factors
# Analyze bookings by country
top_countries = df['country'].value_counts().head(10)
plt.figure(figsize=(10, 6))
top_countries.plot(kind='bar')
plt.title('Top 10 Countries by Number of Bookings')
plt.xlabel('Country')
plt.ylabel('Number of Bookings')
plt.show()

# Analyze bookings by market segment
market_segments = df['market_segment'].value_counts()
plt.figure(figsize=(10, 6))
market_segments.plot(kind='pie', autopct='%1.1f%%')
plt.title('Bookings by Market Segment')
plt.ylabel('')
plt.show()

# 4. Analyze special requests
plt.figure(figsize=(10, 6))
sns.countplot(x='total_of_special_requests', data=df)
plt.title('Distribution of Special Requests')
plt.xlabel('Number of Special Requests')
plt.ylabel('Count')
plt.show()

# Analyze factors influencing special requests
factors = ['adults', 'children', 'babies',
'is_repeated_guest', 'previous_cancellations']
fig, axes = plt.subplots(2, 3, figsize=(18, 12))
axes = axes.flatten()

for i, factor in enumerate(factors):
    sns.boxplot(x=factor, y='total_of_special_requests',
data=df, ax=axes[i])
    axes[i].set_title(f'Special Requests vs {factor}')

plt.tight_layout()
plt.show()
```

```python
# 5. Additional analyses
# Analyze cancellation rate
cancellation_rate = df['is_canceled'].mean() * 100
print(f"Cancellation rate: {cancellation_rate:.2f}%")

# Analyze booking changes
plt.figure(figsize=(10, 6))
sns.countplot(x='booking_changes', data=df)
plt.title('Distribution of Booking Changes')
plt.xlabel('Number of Booking Changes')
plt.ylabel('Count')
plt.show()

# Analyze customer type
customer_types = df['customer_type'].value_counts()
plt.figure(figsize=(10, 6))
customer_types.plot(kind='pie', autopct='%1.1f%%')
plt.title('Distribution of Customer Types')
plt.ylabel('')
plt.show()

# Analyze average daily rate by room type
plt.figure(figsize=(12, 6))
sns.boxplot(x='reserved_room_type', y='adr', data=df)
plt.title('Average Daily Rate by Room Type')
plt.xlabel('Room Type')
plt.ylabel('Average Daily Rate')
plt.show()
```

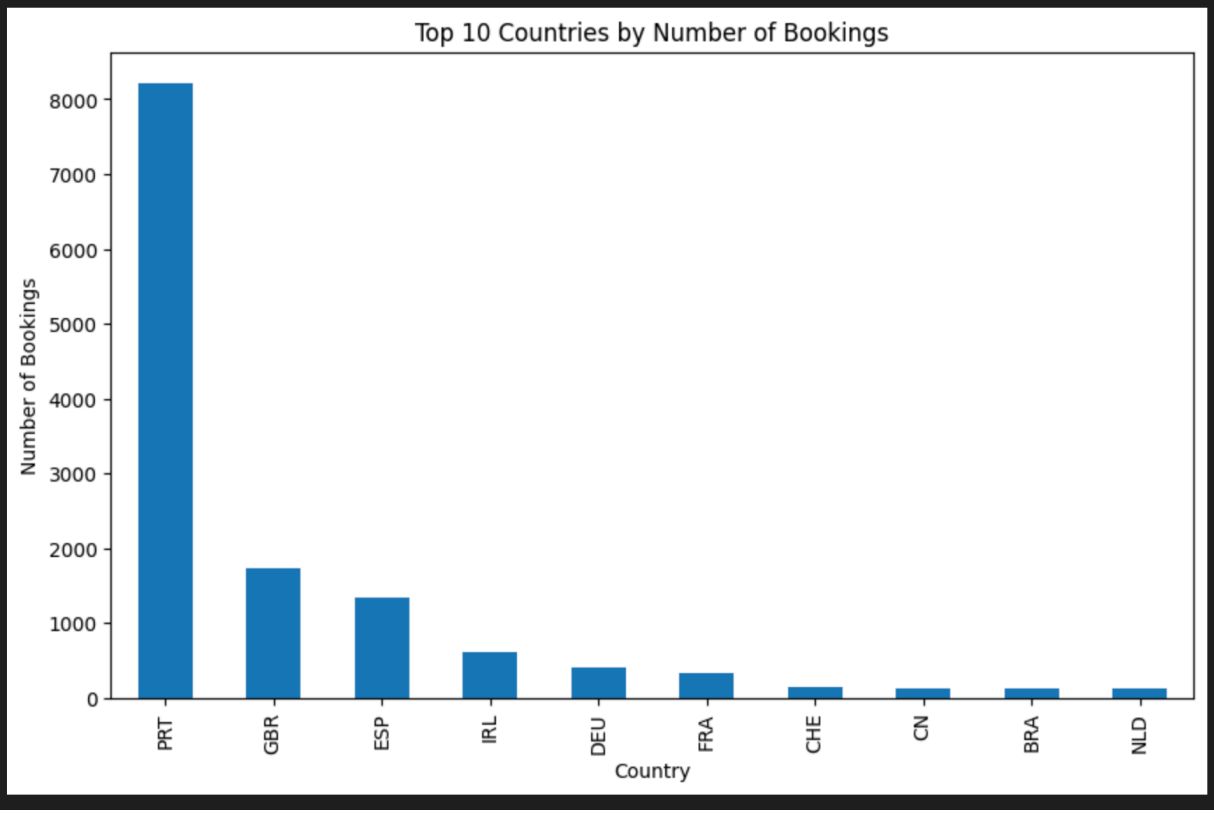
OUTPUT

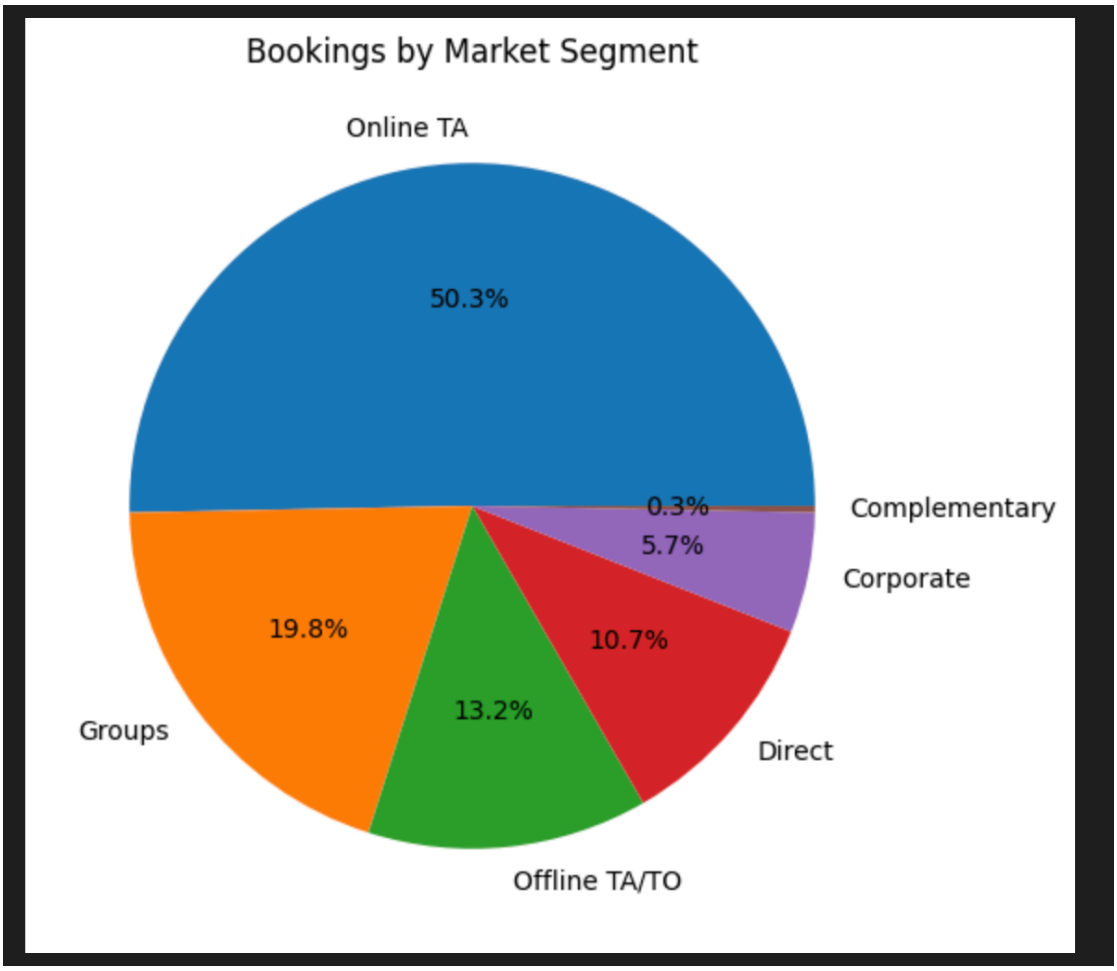

Distribution of Booking Lead Times
Count
Lead Time (days)
0
250
500
750
1000
1250
1500
1750
2000
0
100
200
300
400
500
600
700

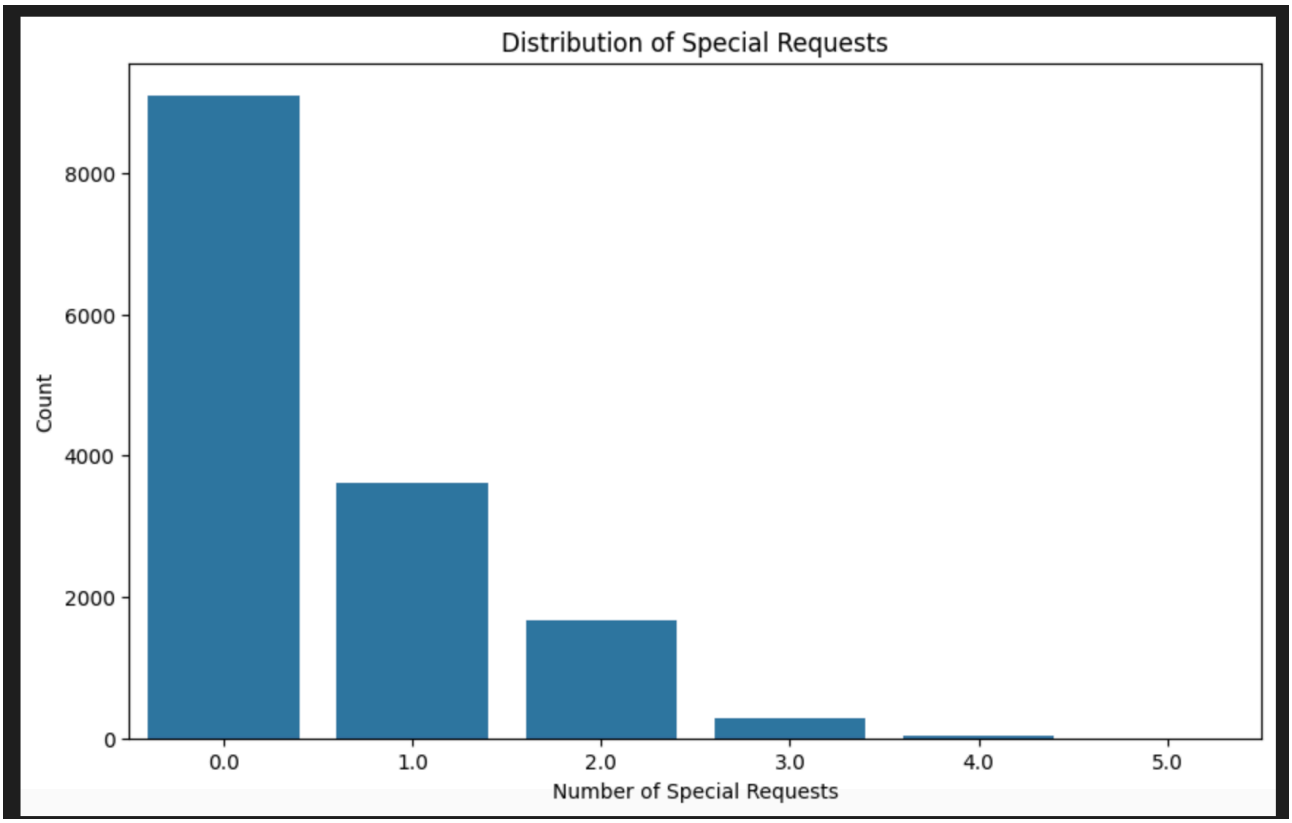

Number of Bookings by Month
Number of Bookings
Month
0
250
500
750
1000
1250
1500
1750
2000
April
August
December
February
January
July
June
March
May
November
October
September

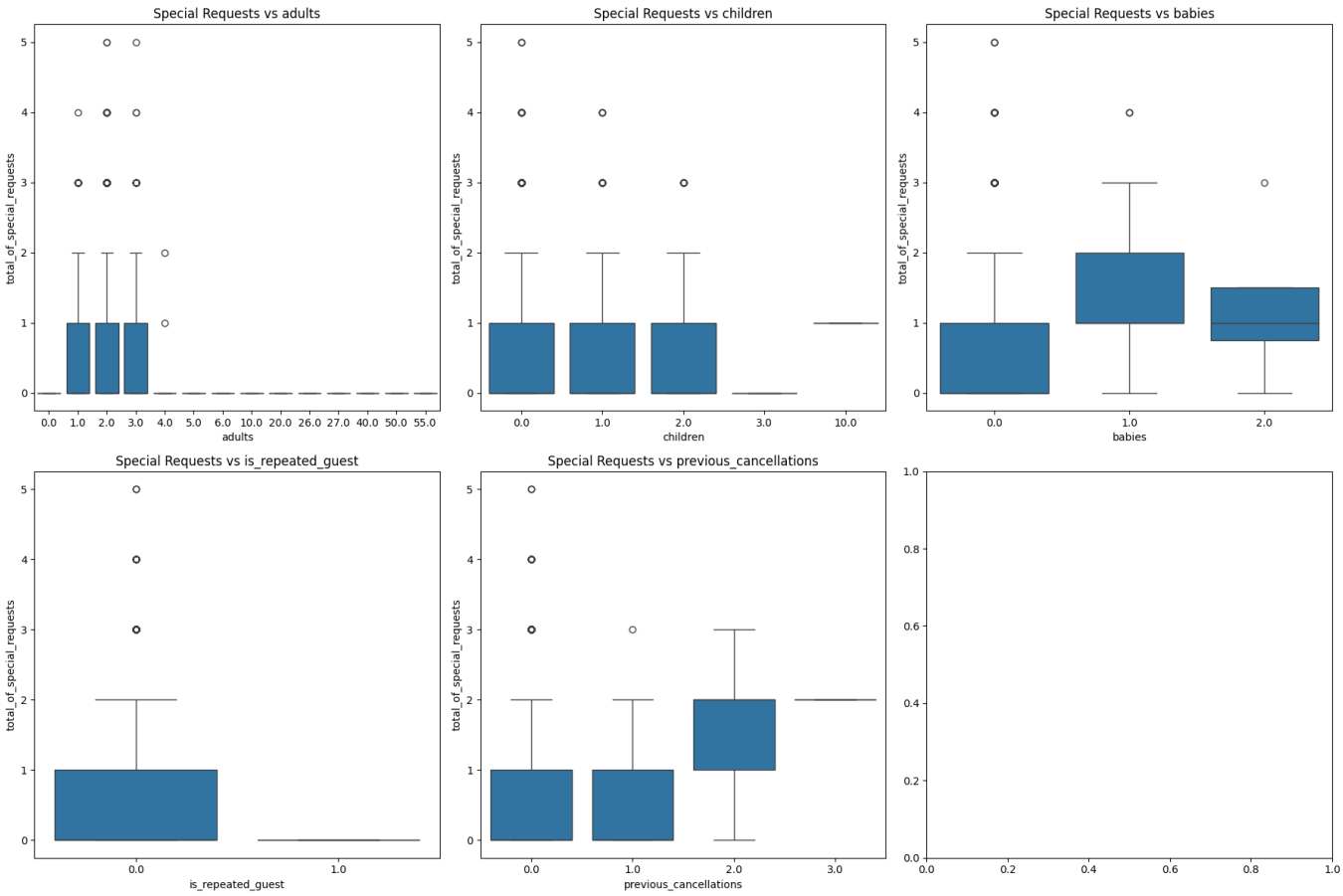


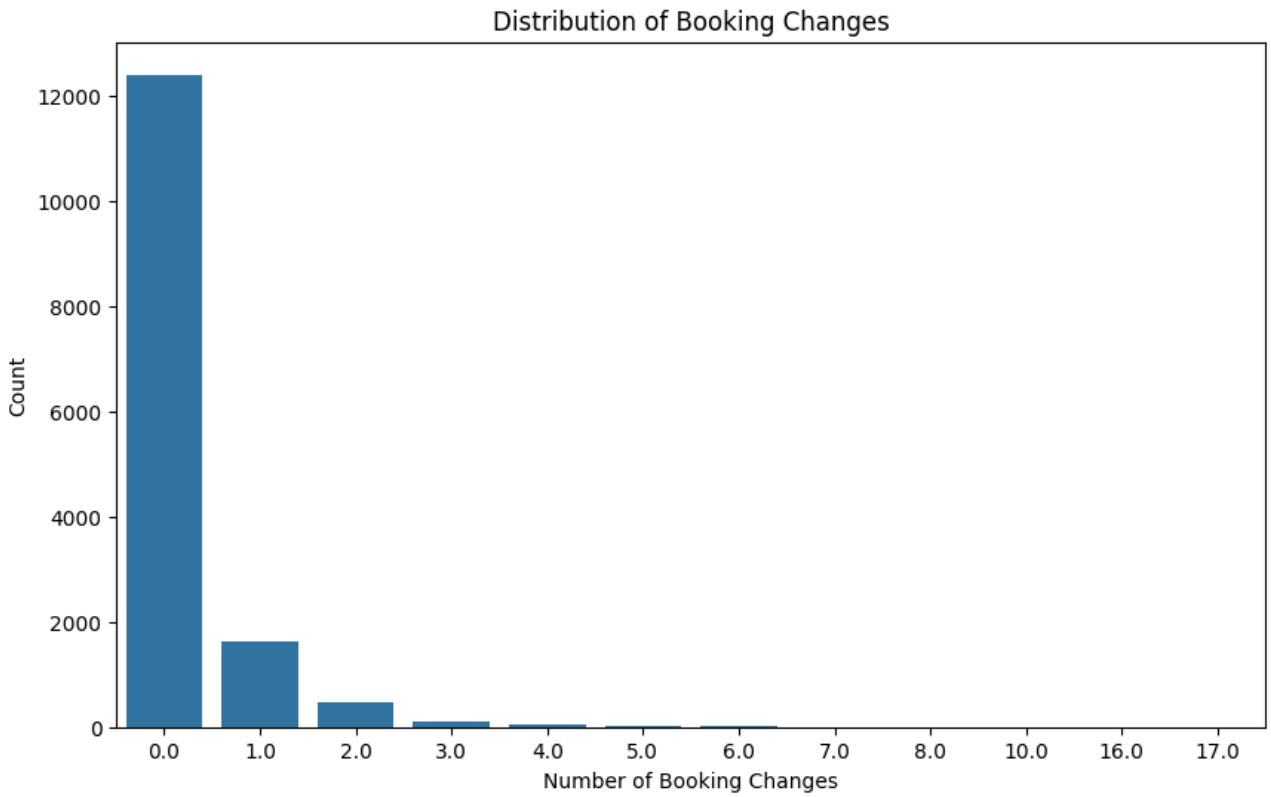

```
Correlation between length of stay and daily rate: 0.11
```

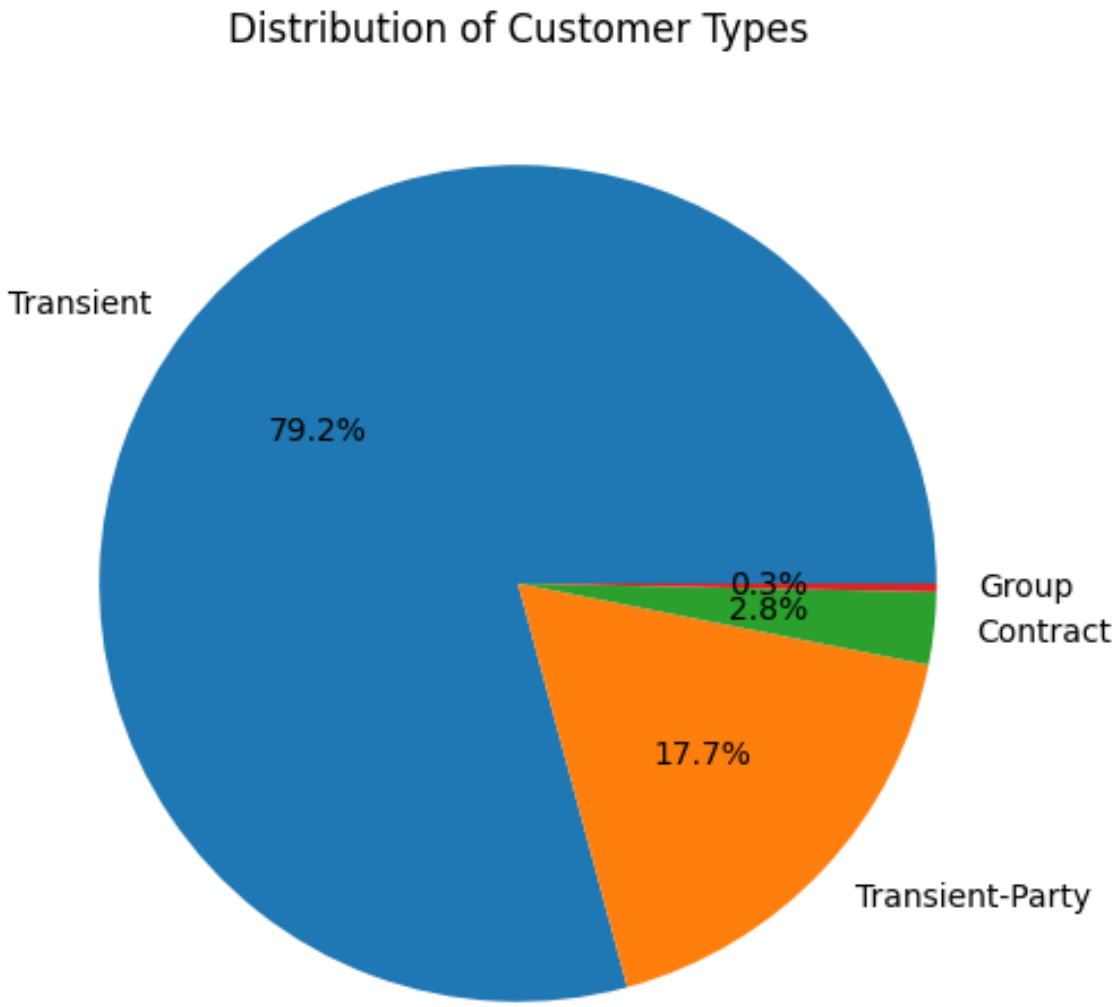


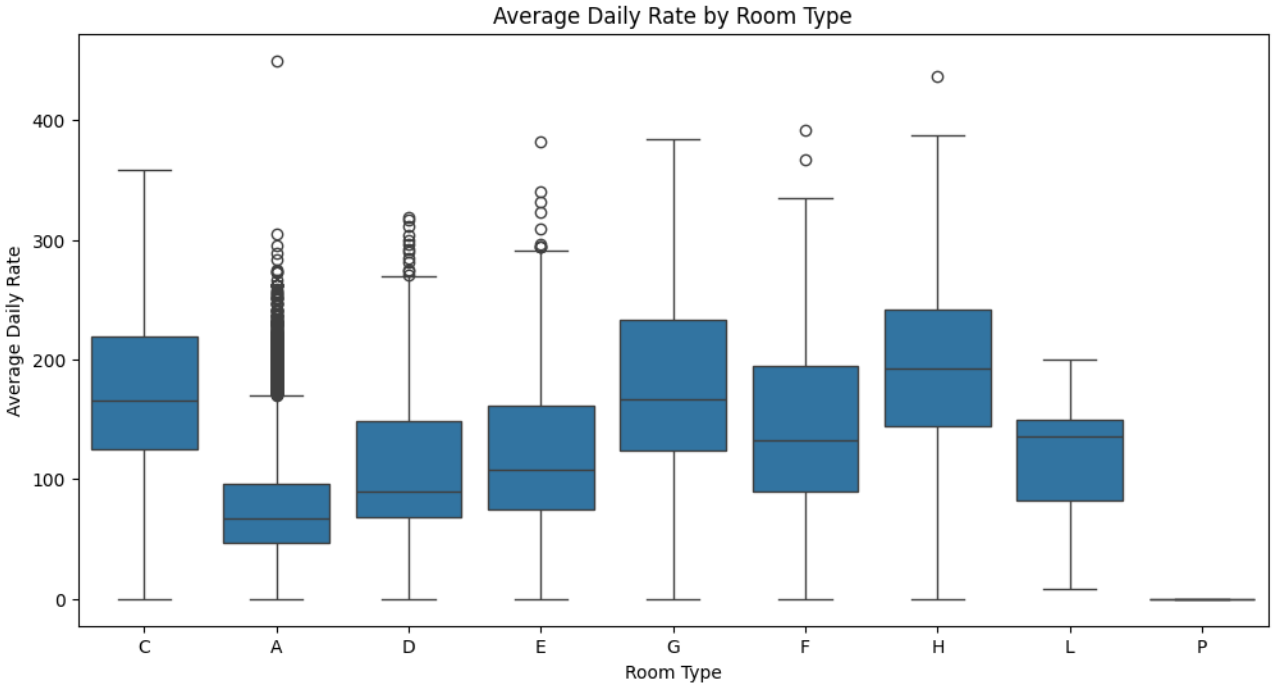

Bookings by Market Segment
Online TA
50.3%
0.3%
Complementary
5.7%
Corporate
19.8%
10.7%
13.2%
Groups
Direct
Offline TA/TO


Distribution of Special Requests
Count
8000
6000
4000
2000
0
0.0
1.0
2.0
3.0
4.0
5.0
Number of Special Requests


Special Requests vs adults
Special Requests vs children
Special Requests vs babies
Special Requests vs is_repeated_guest
Special Requests vs previous_cancellations
total_of_special_requests
adults
children
babies
is_repeated_guest
previous_cancellations


Distribution of Booking Changes
Count
Number of Booking Changes


Distribution of Customer Types
Transient
79.2%
0.3%
2.8%
Group
Contract
17.7%
Transient-Party


Average Daily Rate by Room Type
Average Daily Rate
0
100
200
300
400
C
A
D
E
G
F
H
L
P
Room Type